

Mehmet Akif Duran

mehmetakifdurann@gmail.com | +90 530 084 33 71 | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

EXPERIENCE

Social Media Manager — Meet and Study Library

September 2025 – August 2026

- Grew the library's social media audience by over 2,000 followers and generated 1 million content views across platforms in 11 months.
- Built and maintained a weekly content calendar and promotional campaigns for library events.

Information Technologies Intern — 4S Information Technologies

August 2025 – September 2025

- Installed, administered, and maintained Active Directory, handling user accounts, group policies, and organizational units.
- Configured network infrastructure and troubleshooted connectivity issues across the corporate environment.
- Performed system monitoring and log analysis to identify potential security vulnerabilities.

Research & Development Intern — Feed Protocol

October 2024 – January 2025

- Researched Zero Knowledge Proof systems and authored technical reports for the protocol engineering team.
- Analyzed Verifiable Random Function and RNG mechanisms to evaluate randomness guarantees for on-chain use.
- Supported development of blockchain infrastructure and decentralized applications (dApps).

PROJECTS

MedAssist — Medication Tracking & Caregiver Management App | React Native, TypeScript, Node.js, PostgreSQL, Redis

- Built a cross-platform React Native (TypeScript) client with centralized state management and offline-first caching for medication adherence tracking.
- Designed a normalized PostgreSQL schema covering patients, caregivers, prescriptions, and dose events, with role-based access control linking each caregiver to multiple dependents.
- Engineered a Redis/BullMQ job queue for timezone-aware dose scheduling, dispatching push notifications via FCM and APNs with retry/backoff and missed-dose escalation to caregivers.
- Exposed a RESTful Node.js/Express API secured with JWT authentication, integrating Stripe webhooks with idempotent payment handling for subscription billing.

OptiSched — University Course Planning & Classroom Reservation System | Python, PostgreSQL

- Scheduled 890 lectures across 50 classrooms by modeling timetabling as a constraint-satisfaction problem, using backtracking with forward-checking and graph-coloring heuristics to eliminate instructor, room, and time-slot conflicts.
- Developed a greedy plus local-search optimizer to minimize schedule gaps and balance classroom utilization under capacity and equipment constraints.
- Built a reservation module with transactional conflict detection and database-level triggers preventing real-time double-booking.

FileBridge — File Transfer Application | C++, Multi-threading

- Implemented chunked, resumable transfers with byte-offset tracking and integrity checks, backed by a multi-threaded transfer queue with connection pooling.
- Built a session manager for encrypted credential storage and a drag-and-drop interface reporting real-time progress and throughput.

ECG Signal Analysis & Arrhythmia Detection | C++, MATLAB

- Implemented a DSP pipeline with bandpass filtering, baseline-wander removal, and Pan-Tompkins QRS detection, deriving RR-interval and heart-rate-variability features to flag bradycardia, tachycardia, and irregular rhythm against annotated ECG datasets.

EDUCATION

B.Sc. Computer Engineering — Başkent University, Ankara

September 2023 – January 2027 (Expected)

- Transferred from Istanbul Bilgi University, Computer Engineering (100% English), 2022 – 2023. YÖKDİL Foreign Language Test (English – Natural Sciences): 78.75, August 2026.

LANGUAGES & TECHNOLOGIES

Languages: C, C++, C#, Java, Python, TypeScript, JavaScript, SQL, R, MATLAB

Backend, Data & Mobile: .NET, Node.js, Express, PostgreSQL, Redis, BullMQ, REST APIs, JWT, React Native

Systems & Networks: Multi-threaded and socket programming, TCP/IP, UDP, Active Directory, Docker

Engineering Practice: OOP, SOLID, design patterns, algorithms and data structures, Git

Other: Blockchain and dApps, Zero Knowledge Proofs, Arduino, Proteus

LEADERSHIP

Lead — Google Developer Groups on Campus, Başkent University September 2025 – Present

- Lead a 20-member student developer community, organizing 15 technical events, workshops, and training sessions.

President — Başkent Computer Society, Başkent University October 2024 – October 2025

- Organized technical events and workshops for computer engineering students.